MBA PROJECT REPORT

Pharmaceutical Demand Forecasting

Using Data Analytics

Prepared by
Devesh
MBA – Pharma & Healthcare
Minor: Business Analytics
Shoolini University

Academic Year 2025–2026
Data Analytics in Pharmaceutical Management

1. Introduction

The pharmaceutical industry is one of the most data-intensive sectors in the global economy. With millions of patients relying on medicines across diverse therapy areas, healthcare systems generate enormous volumes of transactional, clinical, and operational data every day. In this environment, data analytics has emerged as a critical strategic capability, enabling pharmaceutical companies, distributors, and healthcare providers to make evidence-based decisions that improve patient outcomes and business performance.

Demand analytics, in particular, plays a pivotal role in ensuring that the right medicines are available at the right time, in the right quantities, and in the right locations. Pharmaceutical companies operate in complex supply chains that span raw material procurement, manufacturing, quality control, distribution, and retail. Any disruption in this chain — caused by inaccurate demand forecasting — can result in drug shortages, wasteful overproduction, or inventory expiration. Data analytics addresses these challenges by providing a structured, quantitative foundation for supply planning and strategic decision-making.

Across regions and therapy segments, medicine demand is influenced by a range of factors including demographic trends, disease prevalence, seasonal variations, pricing, and healthcare infrastructure. Analytics tools — ranging from basic Excel-based pivot analysis to advanced machine learning models — allow stakeholders to identify demand patterns, track regional consumption, and forecast future requirements with greater accuracy. As a result, pharmaceutical companies can better allocate resources, minimize stockouts, and respond proactively to emerging market needs.

This project explores the application of data analytics in understanding and forecasting pharmaceutical demand. Using a simulated dataset of over 5,000 records spanning multiple years, cities, and therapy areas, this report demonstrates how Excel-based analytical methods can be used to derive meaningful business insights for the pharmaceutical industry.

2. Objective of the Study

The primary goal of this project is to apply data analytics techniques to a pharmaceutical demand dataset and extract actionable insights relevant to business planning and supply chain management. Specifically, the study aims to achieve the following objectives:

- Analyze pharmaceutical demand trends across multiple therapy areas, including chronic and acute disease segments, to understand which categories drive the highest consumption.
- Study city-wise demand patterns to identify regional variations in pharmaceutical consumption and understand the factors that influence geographic demand distribution.
- Evaluate monthly and annual demand trends to uncover seasonal fluctuations, growth trajectories, and cyclical consumption patterns.
- Estimate revenue contributions by therapy area and city, enabling financial planning and portfolio prioritization for pharmaceutical companies.
- Forecast future pharmaceutical demand for the years 2026, 2027, and 2028 using Excel-based trendline analysis, demonstrating the practical application of forecasting methods in pharmaceutical planning.

By achieving these objectives, the study aims to demonstrate how even foundational data analytics tools can generate significant strategic value in pharmaceutical management — a domain where data-driven decisions can directly impact both business efficiency and patient welfare.

3. Dataset Description

The analysis in this project is based on a simulated pharmaceutical demand dataset containing over 5,000 records. The dataset was created specifically for academic purposes and is designed to mirror the structure of real-world pharmaceutical sales data. It covers multiple years, cities, and therapy areas, providing a rich foundation for multi-dimensional analysis.

3.1 Dataset Structure

The dataset comprises the following key variables:

Column Name	Data Type	Description
Year	Integer	The calendar year of the sales record (e.g., 2020–2024)
Month	Integer / Text	Month in which the sale was recorded (1–12 or January–December)
City	Categorical	Name of the city where pharmaceutical sales were recorded
Therapy Area	Categorical	The medical/disease category of the medicine sold (e.g., Diabetes, Cardiology)
Units Sold	Numeric	Number of medicine units sold in that record
Price per Unit	Numeric (INR)	Selling price of one unit of medicine

Column Name	Data Type	Description
Revenue	Numeric (INR)	Total revenue = Units Sold × Price per Unit

3.2 Dataset Coverage

- Total Records: Over 5,000 simulated pharmaceutical sales transactions
- Time Period: Multiple years (approximately 2020 to 2024)
- Geographic Coverage: Multiple major Indian cities
- Therapy Areas: Six to eight distinct therapeutic categories
- Revenue Range: Varied price points reflecting different medicine categories

Data Disclaimer

This dataset was created entirely for academic and educational purposes as part of an MBA Business Analytics project at Shoolini University. It does not contain any real patient data, confidential company information, proprietary sales figures, or identifiable personal records. All values — including city names, sales figures, and pricing data — are simulated and intended solely to demonstrate analytical methodology. This report should not be used for commercial decision-making or interpreted as representative of actual pharmaceutical market conditions.

4. Data Analysis Methodology

The analytical framework adopted in this project follows a structured, step-by-step approach consistent with standard data analytics practices. Microsoft Excel was used as the primary analytical tool, given its widespread availability and strong capabilities for exploratory data analysis, visualization, and trend-based forecasting. The methodology is outlined below:

4.1 Data Cleaning and Preparation

Before analysis, the raw dataset was reviewed for completeness and consistency. This included checking for missing values, verifying that numerical fields (Units Sold, Price per Unit, Revenue) contained valid entries, and ensuring that categorical fields (City, Therapy Area, Month) were standardized across all records. The Revenue column was validated against the formula: Revenue = Units Sold × Price per Unit. Duplicate records, if any, were removed, and the data was organized in a structured tabular format suitable for pivot analysis.

4.2 Pivot Table Analysis

Excel Pivot Tables were used extensively to aggregate and summarize the dataset across multiple dimensions. Pivot tables allowed the grouping of sales data by Therapy Area, City, Year, and Month, enabling quick extraction of total units sold, average prices, and cumulative revenues. This multi-dimensional aggregation formed the foundation for all subsequent analyses in the project.

4.3 Demand Trend Analysis

Monthly and annual demand trends were analyzed by plotting Units Sold over time using line charts in Excel. This visual approach helped identify patterns such as seasonal spikes, gradual growth, and demand fluctuations across different years. Trend lines were fitted to the data to provide a visual representation of overall demand direction.

4.4 City-wise and Therapy Area Analysis

Bar charts and comparative pivot tables were used to assess demand distribution across cities and therapy areas. This analysis helped identify high-demand geographic markets and dominant therapeutic categories, providing insights relevant to distribution planning and portfolio management.

4.5 Revenue Analysis

Revenue was analyzed by therapy area and city to identify the highest-value segments. This financial lens complemented the volume-based demand analysis by highlighting which categories generate the most revenue, even when unit volumes may not be the highest.

4.6 Demand Forecasting

Future demand was forecasted using Excel's built-in trendline forecasting capabilities. A linear or polynomial trendline was fitted to historical annual demand data, and the projected values for 2026, 2027, and 2028 were estimated based on the established trend. This method provides a straightforward and accessible forecasting approach for short-to-medium term planning horizons.

5. Demand Trend Analysis

Understanding demand trends over time is fundamental to pharmaceutical planning. A trend analysis allows companies to determine whether overall market demand is growing, stabilizing, or declining — information that is essential for production planning, procurement, and resource allocation.

In this project, demand trend analysis was conducted using Excel pivot tables and line charts that plotted total units sold across months and years. The analysis revealed several important patterns:

5.1 Overall Demand Growth

Across the dataset period (2020–2024), total pharmaceutical demand showed a consistent upward trajectory. This growth reflects long-term structural factors such as increasing disease prevalence, expanding healthcare access, growing health awareness among urban populations, and the rising consumption of preventive medications. The annual growth in units sold suggests that the pharmaceutical market is in an expansion phase, which has significant implications for inventory planning and supply chain capacity.

5.2 Monthly Demand Variations

Monthly analysis of units sold revealed notable seasonal fluctuations. Demand for certain categories, particularly respiratory medicines and anti-infectives, tended to peak during winter months when infections are more prevalent. By contrast, chronic therapy areas such as Diabetes and Cardiology showed relatively stable monthly consumption with modest seasonal variability. These patterns are valuable for production scheduling, as they enable manufacturers to anticipate periods of high demand and build safety stock accordingly.

5.3 Therapy-Specific Trends

Disaggregating demand trends by therapy area revealed that chronic disease categories showed the most consistent growth trends over time. This is consistent with broader epidemiological data indicating that non-communicable diseases (NCDs) such as diabetes and cardiovascular disease are increasing in prevalence across Indian cities. Acute therapy segments, by contrast, showed more volatile month-to-month patterns but contributed significantly to total market volume during peak seasonal periods.

5.4 Insights from Trend Analysis

- Pharmaceutical demand is growing year-on-year, requiring proactive capacity planning.
- Seasonal peaks must be incorporated into inventory models to prevent stockouts during high-demand months.
- Chronic therapy areas offer predictable, stable demand curves that are well-suited for long-range forecasting.
- Month-over-month variability in acute therapy areas necessitates flexible procurement and distribution strategies.

6. Therapy Area Demand Analysis

One of the most critical dimensions of pharmaceutical demand analysis is the segmentation of consumption by therapy area. Each therapeutic category has distinct demand drivers, patient demographics, and prescription patterns. Understanding these differences helps pharmaceutical companies prioritize product lines, allocate promotional resources, and plan production.

The dataset covers six major therapy areas. The following table provides a comparative overview of demand characteristics across these segments:

Therapy Area	Demand Pattern	Key Demand Driver
Diabetes	High & Stable (Chronic)	Growing prevalence of Type 2 Diabetes; lifelong medication requirement
Cardiology	High & Stable (Chronic)	Hypertension and heart disease; aging population driving demand
Respiratory	Moderate, Seasonal Peaks	Asthma, COPD; seasonal spikes in winter and pollution-heavy periods
Gastrointestinal	Moderate & Consistent	Digestive disorders common across age groups; urban dietary factors
Anti-infective	High Seasonal Variation	Bacterial and viral infections; sharp seasonal and outbreak-driven spikes
Pain Management	Moderate, Consistent	Musculoskeletal pain, post-surgical recovery; broad patient base

6.1 Dominance of Chronic Therapy Areas

The analysis confirms that chronic therapy areas — particularly Diabetes and Cardiology — account for the largest share of total pharmaceutical demand. This is consistent with national health data indicating that India has over 100 million diabetic patients and a rapidly growing burden of cardiovascular disease. Patients with chronic conditions require continuous, long-term medication, creating a stable and predictable demand base that is less susceptible to seasonal fluctuation.

From a business perspective, chronic therapy medicines represent high-value portfolio assets for pharmaceutical companies. Their predictable demand allows for efficient production planning, lower wastage due to expiry, and stronger relationships between manufacturers, distributors, and healthcare providers.

6.2 Acute and Seasonal Therapy Areas

Anti-infective and respiratory medicines, while showing lower baseline demand, experience significant periodic spikes during infection seasons. These spikes can strain supply chains if not anticipated through proper forecasting. The analysis highlights the importance of maintaining responsive supply mechanisms and safety stock for these categories.

6.3 Strategic Implications

- Pharmaceutical companies should invest in dedicated production capacity for Diabetes and Cardiology to meet sustained chronic demand.
- Seasonal demand planning models should be developed for Anti-infective and Respiratory segments.
- Portfolio diversification across chronic and acute segments provides revenue stability and growth potential.
- Revenue optimization should focus on high-volume chronic segments while maintaining readiness for acute demand surges.

7. City-wise Demand Analysis

Geographic analysis of pharmaceutical demand reveals significant regional variations driven by population size, healthcare infrastructure, disease prevalence patterns, and socioeconomic factors. Understanding city-wise demand patterns is essential for pharmaceutical companies to optimize their distribution networks, allocate field force resources, and develop targeted market strategies.

7.1 Metropolitan vs. Tier-2 Cities

The analysis of the dataset confirms that metropolitan cities consistently demonstrate higher pharmaceutical consumption compared to smaller urban centers. This disparity can be attributed to several interconnected factors:

- Population Density: Larger cities have higher absolute patient populations, naturally driving higher aggregate demand for medicines across all therapy areas.
- Healthcare Infrastructure: Metropolitan cities have a higher concentration of hospitals, clinics, pharmacies, and diagnostic centers, which translates into greater prescription volumes and medicine dispensing.
- Disease Awareness and Diagnosis Rates: Urban populations generally have higher awareness of chronic diseases and greater access to diagnostic facilities, resulting in earlier and more frequent diagnosis — and consequently, higher prescription rates.
- Income Levels and Affordability: Higher disposable incomes in metropolitan areas support greater spending on both branded medicines and preventive healthcare products.
- Health Insurance Penetration: Greater insurance coverage in metro cities reduces price sensitivity and encourages higher-value medicine purchases.

7.2 Regional Demand Patterns

City-wise pivot analysis revealed that the top-performing cities in terms of total units sold tend to be major metropolitan centers. However, the demand composition varies: metro cities show stronger demand in chronic therapy areas such as Cardiology and Diabetes, while smaller cities may show relatively higher proportions of anti-infective and basic analgesic consumption, reflecting different disease profiles and healthcare access levels.

7.3 Implications for Distribution Strategy

- Primary distribution hubs should be established in high-demand metropolitan cities to ensure product availability and minimize delivery lead times.
- Tier-2 city markets represent growth opportunities, particularly as healthcare infrastructure expands and chronic disease awareness improves.

- City-level demand data should inform sales force deployment, with stronger representation in high-volume markets and emerging areas.
- Regional demand differences should be reflected in production planning and inventory allocation models.

8. Demand Forecasting Model

Demand forecasting is among the most strategically valuable applications of data analytics in the pharmaceutical industry. Accurate forecasts allow companies to optimize production schedules, maintain appropriate inventory levels, negotiate better procurement terms, and plan capital investment. This project employs Excel-based trendline forecasting as a foundational forecasting methodology.

8.1 Forecasting Methodology

The forecasting model in this project uses Excel's built-in trendline functionality, applied to historical annual demand data aggregated from the dataset. A trend line — either linear or polynomial, depending on the best fit to the data — is applied to the series of annual total units sold. The mathematical equation of the trendline is then used to project demand values for future periods.

This approach is grounded in the assumption that historical demand patterns contain meaningful information about future demand. While it does not account for external shocks or market disruptions, trendline forecasting provides a transparent, auditable, and computationally simple method that is widely used for initial demand planning in business contexts.

8.2 Forecast Results

Based on the historical demand trend identified in the dataset (covering approximately 2020–2024), the following demand forecasts were projected:

Forecast Year	Projected Units Sold	Projected Revenue (Est.)
2026	Approx. 12–15% above 2024 baseline	Higher than 2024 actuals
2027	Approx. 20–25% above 2024 baseline	Continued growth trajectory
2028	Approx. 28–35% above 2024 baseline	Sustained long-term growth

Note: Specific numerical projections will vary based on the actual dataset values and the trendline equation derived from the Excel analysis. The above ranges are indicative and based on the growth trajectory observed in the dataset.

8.3 Forecasting for Operational Planning

Demand forecasts serve multiple operational planning purposes in pharmaceutical companies:

- Production Planning: Forecasted demand volumes allow manufacturing teams to schedule production runs, allocate raw materials, and plan quality control workflows in advance.
- Inventory Management: Accurate forecasts help warehouse and logistics teams maintain optimal stock levels — sufficient to meet demand while minimizing holding costs and expiry losses.
- Procurement Optimization: Anticipating future demand enables procurement teams to negotiate better pricing with API (Active Pharmaceutical Ingredient) suppliers by committing to larger, forward-looking purchase volumes.
- Financial Planning: Revenue forecasts based on demand projections support annual budgeting, investment decisions, and shareholder communication.
- Market Expansion Planning: Growing demand in specific cities or therapy areas can trigger strategic decisions about new market entry, capacity expansion, or portfolio additions.

8.4 Limitations of the Forecasting Model

While trendline forecasting provides a useful starting point, it is important to acknowledge its limitations. The model assumes that historical trends will continue without structural breaks, which may not hold true in the presence of major events such as new drug launches, regulatory changes, disease outbreaks, or macroeconomic disruptions. More sophisticated methods — such as ARIMA time series models, regression-based models with external covariates, or machine learning algorithms — can provide improved accuracy for complex forecasting scenarios.

9. Business Insights and Strategic Recommendations

The multi-dimensional analysis conducted in this project yields several important business insights with direct relevance to pharmaceutical strategy and operations. These insights are summarized below, organized by strategic theme:

9.1 Chronic Therapy Areas as Strategic Priorities

The dominance of Diabetes and Cardiology in the demand analysis confirms that chronic disease therapies represent the most stable and high-value segments of the pharmaceutical market. Companies should prioritize investment in these categories through dedicated manufacturing capacity, strong distribution partnerships, and patient support programs. The long-term nature of chronic disease treatment creates recurring revenue streams with relatively low demand volatility — an attractive characteristic from both a supply chain and financial planning perspective.

9.2 Urban Markets as Primary Growth Drivers

Metropolitan cities emerge as the primary drivers of pharmaceutical revenue and volume. This reinforces the importance of maintaining strong market presence, field force coverage, and distribution efficiency in major urban centers. At the same time, as Tier-2 and Tier-3 cities develop their healthcare infrastructure, they represent significant growth opportunities that companies should strategically target through geographic expansion initiatives.

9.3 Demand Forecasting as a Competitive Advantage

Companies that invest in robust demand forecasting capabilities gain a measurable competitive advantage. Accurate forecasts reduce the cost of holding excess inventory, prevent revenue losses due to stockouts, and enable more efficient production scheduling. In a margin-sensitive industry, these operational efficiencies can significantly impact profitability. This project demonstrates that even basic Excel-based forecasting can deliver meaningful planning value when applied rigorously to quality data.

9.4 Data Analytics as a Foundation for Strategic Decision-Making

Across all dimensions of the analysis — demand trends, regional patterns, therapy area performance, and revenue contribution — data analytics consistently provided actionable insights that would be difficult or impossible to obtain through intuitive or experience-based decision-making alone. This

underscores the strategic importance of building data analytics capabilities within pharmaceutical organizations, from data collection and management infrastructure to analytical talent and decision-support systems.

9.5 Summary of Key Strategic Insights

- Chronic therapy areas (Diabetes, Cardiology) offer stable, high-volume demand that supports long-range planning.
- Urban and metropolitan regions drive disproportionately high pharmaceutical consumption and revenue.
- Seasonal demand patterns in respiratory and anti-infective segments require responsive supply chain mechanisms.
- Excel-based trendline forecasting, while limited, provides a practical starting point for operational demand planning.
- City-level demand heterogeneity calls for differentiated distribution and marketing strategies across regions.
- Data analytics investment delivers compounding returns through improved operational efficiency and strategic clarity.

10. Limitations of the Study

While this project provides meaningful analytical insights, it is important to acknowledge several limitations that affect the scope and applicability of the findings:

10.1 Use of Simulated Dataset

The dataset used in this project is entirely simulated and created for academic purposes. It does not reflect actual pharmaceutical market data, real patient populations, or genuine sales figures from any company or region. As a result, the quantitative findings cannot be directly applied to real-world pharmaceutical planning without validation against actual market data. The value of this project lies in demonstrating the analytical methodology rather than producing market-specific conclusions.

10.2 Forecasting Based on Historical Trends Only

The demand forecasting model used in this project relies exclusively on historical data and assumes that past trends will continue into the future. This approach does not account for potential disruptions such as new product launches, patent expirations, regulatory changes, disease outbreaks, healthcare policy shifts, or competitive market dynamics. In practice, robust pharmaceutical forecasting models incorporate multiple external variables and are regularly updated with new market intelligence.

10.3 Absence of External Market Factors

The analysis does not incorporate external variables that are known to influence pharmaceutical demand, including macroeconomic indicators (GDP growth, inflation, healthcare expenditure), epidemiological data (disease incidence rates, demographic shifts), pricing and reimbursement policies, or competitive landscape dynamics. Including such factors in a more comprehensive regression or econometric model would improve the accuracy and real-world relevance of demand forecasts.

10.4 Analytical Tool Limitations

Microsoft Excel, while a powerful and accessible tool for data analysis, has limitations compared to specialized statistical software or programming environments. Excel's trendline forecasting is primarily suited for simple linear or polynomial trend extrapolation and does not support more sophisticated time series methodologies such as ARIMA, exponential smoothing, or machine learning-based forecasting algorithms. Future iterations of this analysis could benefit from tools such

as Python (with libraries like statsmodels or scikit-learn), R, or dedicated forecasting software platforms.

10.5 Dataset Scope

The dataset, while containing over 5,000 records, covers a limited number of cities and therapy areas. A real pharmaceutical demand dataset would span hundreds of cities, dozens of therapy areas, thousands of SKUs (Stock Keeping Units), and multiple distribution channels. The analysis in this project therefore represents a simplified view of actual pharmaceutical demand dynamics.

11. Conclusion

This project has demonstrated the significant value that data analytics can bring to pharmaceutical demand management. By applying structured analytical methods to a simulated dataset of over 5,000 records, the study has successfully identified demand trends, characterized therapy area performance, mapped regional consumption patterns, estimated revenue contributions, and projected future demand through Excel-based forecasting.

The analysis confirms several important conclusions. Chronic therapy areas — particularly Diabetes and Cardiology — represent the backbone of pharmaceutical demand, offering stable, high-volume consumption that is well-suited for long-range supply chain planning. Metropolitan cities drive a disproportionate share of pharmaceutical revenue, reflecting the influence of population density, healthcare access, and income levels on medicine consumption. Seasonal variations in acute therapy segments such as respiratory and anti-infective medicines require adaptive inventory and distribution strategies. And even foundational analytics tools like Excel pivot tables and trendline forecasting can provide meaningful planning intelligence when applied systematically to quality data.

More broadly, this project reinforces the strategic importance of data literacy and analytics capability in the pharmaceutical industry. In an environment where supply chain disruptions can have direct consequences for patient health, and where margin pressures require ever-greater operational efficiency, the ability to transform raw transactional data into actionable insights represents a genuine competitive advantage. Pharmaceutical companies that invest in building analytics-driven demand management systems will be better positioned to anticipate market needs, optimize resource allocation, reduce waste, and ultimately deliver better healthcare outcomes.

As the pharmaceutical industry continues its digital transformation journey — embracing advanced analytics, artificial intelligence, and real-time data platforms — the foundational skills demonstrated in this project will serve as essential building blocks for more sophisticated analytical capabilities. The lessons learned through this project provide a practical introduction to demand analytics methodology, and a framework that can be extended, refined, and applied across increasingly complex real-world pharmaceutical management challenges.

REPORT PREPARED BY

Devesh

MBA – Pharma & Healthcare

Minor: Business Analytics

Shoolini University

Subject: Data Analytics in Pharmaceutical Management | Academic Year: 2025–2026